

SALES TRANSACTION ANALYTICS DASHBOARD

1. Project Overview

The **Sales Transaction Analytics Dashboard** is a Power BI project developed to analyze sales performance and transaction activity. The dashboard transforms sales transaction data into interactive KPIs and visualizations covering sales performance, transaction volume, voucher types, monthly trends, quarterly distribution, and yearly performance.

2. Project Objectives

- Analyze total sales.
- Monitor the number of transactions.
- Calculate average sales.
- Identify minimum and maximum sales values.
- Analyze monthly sales trends.
- Compare sales by voucher type.
- Analyze quarterly sales distribution.
- Compare yearly sales performance.
- Analyze transactions by voucher type.
- Provide interactive filtering through slicers.

3. Data Model

SalesData: Single-table model containing sales transaction records and fields including Date, Month, Quarter, Year, Voucher Type, and Sales/Transaction Amount.

4. Key Performance Indicators

KPI	Purpose
Total Sales	Total sales amount.
Total Transactions	Total number of sales transactions.
Average Sales	Average sales value.
Minimum Sale	Lowest recorded sales value.
Maximum Sale	Highest recorded sales value.

5. Dashboard Visualizations

- Monthly Sales Trend – line chart showing sales performance across months.
- Sales by Voucher Type – clustered bar chart comparing sales across voucher types.
- Quarter-wise Sales Distribution – donut chart showing quarterly sales distribution.
- Year-wise Sales – clustered column chart comparing yearly performance.
- Date-based Sales Trend – area chart showing sales trends using transaction Date.
- Transactions by Voucher Type – clustered column chart showing transaction counts by voucher type.

6. Interactive Slicers

- Year
- Month
- Quarter
- Voucher Type

7. Tools and Technologies

- Microsoft Power BI
- Power Query
- DAX
- Data Visualization
- Data Analysis
- Excel/Tally POS Data

8. Data Quality Consideration

The Month, Quarter, and Year fields should be checked against the verified **Date** field before business reporting. If these fields were manually generated or exported from a source system, discrepancies may occur between them and the actual transaction Date.

For improved reliability, future versions should derive Year, Quarter, Month, and Month Number directly from the verified Date field or from a dedicated Calendar table.

9. Business Insights

- Overall sales performance.
- Transaction volume.
- Monthly sales movements.
- Voucher-type performance.
- Quarterly sales distribution.
- Yearly sales performance.
- Transaction concentration across voucher types.

10. Data Analysis Approach

- Import sales transaction data.
- Review and clean the dataset.
- Validate the Date field.
- Create required DAX measures.
- Build KPI cards.
- Create monthly, quarterly, and yearly visualizations.
- Add voucher-type analysis.
- Add interactive slicers.
- Validate the dashboard against source data.

- Publish the completed Power BI dashboard.

11. Conclusion

The Sales Transaction Analytics Dashboard demonstrates the practical application of Power BI for sales reporting and business intelligence. The project combines KPI monitoring, time-based analysis, voucher-type analysis, transaction analysis, interactive filtering, and data-quality validation to provide a consolidated view of sales performance.